

Day – 13 Troubleshooting Network services practical.

Task – 1 DNS resolution failure.

- Even though I am running in the cloud, you can easily test this exact scenario on your own machine. Here is how to perform the simulation, along with the exact error messages you will observe:
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- Step 1: Edit Your Local hosts File
- To force your computer to map www.spotify.com to an incorrect IP address (for example, 192.168.1.99), open your system's hosts file with administrator/root privileges:
- On Windows:
- Open Notepad as Administrator.
- Open the file: C:\Windows\System32\drivers\etc\hosts
- On Linux / macOS:
- Open your terminal and run: `sudo nano /etc/hosts`
- Add the following line at the bottom of the file, save, and close it:
- 192.168.1.99 www.spotify.com
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- Step 2: Test with the ping Command
- Open your command prompt or terminal and run a ping against the domain:
- DOS
- `ping www.spotify.com`
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- Step 3: Documented Output & Error Message
- Depending on your operating system and network setup, you will see output similar to this:
- On Windows:
- Pinging www.spotify.com [192.168.1.99] with 32 bytes of data:
- Request timed out.
- Request timed out.
- Request timed out.
- Request timed out.
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- Ping statistics for 192.168.1.99:

- Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
- On Linux / macOS:
- PING www.spotify.com (192.168.1.99) 56(84) bytes of data.
- --- www.spotify.com ping statistics ---
- 4 packets transmitted, 0 received, 100% packet loss, time 3058ms
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- The Override: When you type www.spotify.com, your operating system checks the hosts file *before* ever querying a public or local DNS server (like 8.8.8.8 or BIND9). It immediately finds the match pointing to 192.168.1.99.
- The Result: Technically, the "resolution" succeeds because the OS found an IP address. However, because 192.168.1.99 is a dead or incorrect endpoint on your local network, your computer sends out ICMP echo requests into the void, resulting in 100% packet loss and timeouts.

Task – 2 Config the DHCP server in network.

- Here is how you can set up, identify, and resolve this exact scenario in a network simulator like Cisco Packet Tracer:
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- Step 1: Intentionally Misconfigure the DHCP Server
- Open your DHCP server configuration in Packet Tracer.
- Go to the Services tab > DHCP.
- In your active pool settings, enter an incorrect Default Gateway address (for example, typing 192.168.1.99 instead of your router's actual gateway address of 192.168.1.1).
- Save the configuration and ensure the DHCP service is turned ON.
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- Step 2: Connect a Client and Verify the Assigned Gateway
- Connect a client PC to the network switch and switch its IP configuration from Static to DHCP.
- Once it completes the DORA handshake, open the PC's Command Prompt and run:
- DOS
- ipconfig /all
- Observe the result: You will see that the client successfully received an IP address (e.g., 192.168.1.10) and a subnet mask (255.255.255.0), but the Default Gateway points incorrectly to 192.168.1.99.
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- Step 3: Identify the Issue (Symptoms & Diagnostics)

- Local Traffic Works: If you have another device on the exact same subnet (192.168.1.x), the client can ping and communicate with it successfully because local traffic doesn't require a gateway.
- External Traffic Fails: If you try to ping an external IP address (like Google's public DNS at 8.8.8.8) or try to browse a domain name, the ping will fail with "Destination Host Unreachable" or request timeouts.
- The Diagnostic Tool: Running a traceroute (tracert 8.8.8.8) on the client will show that packets immediately get stuck trying to reach the non-existent or wrong gateway (192.168.1.99).
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- Step 4: Fix the Misconfiguration
- To correct the issue and restore full network connectivity:
- Fix the Server Pool: Go back to your DHCP server settings, update the Default Gateway field to the correct address (192.168.1.1), and click Save.
- Force Lease Renewal on the Client: Because the client already holds an active lease with the old gateway, you must clear it. Open the client PC's command prompt and run:
 - DOS
 - ipconfig /release
 - ipconfig /renew
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- Final Verification: Run ipconfig /all again to confirm the client has picked up the correct gateway (192.168.1.1), then test with a ping to 8.8.8.8 to ensure external connectivity is fully restored!

Task – 3 Based on scenario.

- Here is the step-by-step troubleshooting workflow using nslookup and dig to diagnose and fix the issue.
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- Step 1: Verify Client-Side DNS Configuration
- Before running advanced diagnostic tools, check what DNS server address your client device is actually assigned to use.
- The Command:
 - DOS
 - ipconfig /all
- What to look for: Locate the DNS Servers entry. If it is blank, points to an old decommissioned server, or lists an incorrect IP address (like 0.0.0.0), the client has no valid resolver to translate domain names.

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- Step 2: Diagnose the Failure Using nslookup
- Test how the client's default resolver handles a domain name lookup.
- The Command:
- DOS
- nslookup www.musicmania.in
- Observed Error Message: You will likely see a response like:
- Server: UnKnown
- Address: 192.168.1.99
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- DNS request timed out.
- timeout was 2 seconds.
- *** UnKnown can't find www.musicmania.in: Server failure
- What this tells you: The client is trying to query 192.168.1.99, but that server is either offline, unreachable, or not configured to answer queries for that zone.
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- Step 3: Isolate the Problem Using Public Resolvers
- To determine whether the issue lies with your local DNS server or the client itself, force nslookup or dig to query a known public DNS server (such as Google's 8.8.8.8 or Cloudflare's 1.1.1.1).
- Using nslookup:
- DOS
- nslookup www.google.com 8.8.8.8
- Using dig (if on Linux/macOS):
- Bash
- dig @8.8.8.8 www.google.com
- Interpreting the Result:
- If this succeeds and returns an IP: Your client's network stack is fine, but your *local* network's primary DNS server (configured via DHCP or statically) is broken.
- If this fails: Your client has a broader network or firewall issue blocking UDP/TCP port 53 traffic entirely.
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- Step 4: Check Local DNS Server Reachability
- If the local DNS server is supposed to be handling your network's requests, check if it is reachable on the network.
- The Command:
- DOS
- ping 192.168.1.50
- What to check: Ensure the DNS server is powered on, has an active IP lease, and that its BIND or DNS service daemon is actively running.

Task – 4 IP is not receiving from DHCP server.

- Possible Causes for APIPA Fallback
- Physical or Link-Layer Disconnection: A loose Ethernet cable, a disabled network adapter, or a misconfigured switch port (such as being assigned to the wrong VLAN) prevents broadcast traffic from reaching the server.
- DHCP Pool Exhaustion: The server has active leases for every single IP address in its pool and has no remaining addresses to allocate to new clients.
- DHCP Server Service Failure: The DHCP server application or service daemon has crashed, stopped running, or the server itself is offline.
- Missing DHCP Relay Agent (ip helper-address): If the client and the DHCP server reside on different subnets or VLANs, the initial broadcast DHCPDISCOVER message cannot cross router boundaries without a configured helper address on the default gateway.
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- Step-by-Step Troubleshooting and Resolution Process
- Step 1: Check Physical and Link-Layer Connectivity
- Action: Inspect the client device's physical link status. Check if the network interface card (NIC) is enabled and verify that link lights on the switch or router are active.
- Verification: In a simulation tool like Cisco Packet Tracer, ensure the link lines are green and that the port isn't shut down or isolated by a mismatched VLAN configuration.
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- Step 2: Verify DHCP Server Operational Status
- Action: Log into the designated DHCP server machine or check its management interface.
- Verification: Ensure the hardware is powered on and that the DHCP Service is actively running (not paused or stopped).
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- Step 3: Inspect the DHCP Scope and Address Pools
- Action: Review the active pool settings on the server. Look at the total lease table to see how many addresses are currently assigned.
- Verification: If all available addresses are claimed (pool exhaustion), either increase the range of the subnet pool or reduce the DHCP lease time so stale, unused leases expire faster.
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- Step 4: Check Cross-Subnet Routing (If Applicable)
- Action: If the client and server are on separate subnets, inspect the default gateway router configuration.

- Verification: Ensure the router interface facing the client has the ip helper-address [DHCP-Server-IP] command configured so it can successfully forward broadcast discovery requests as unicast packets to the server.
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- Step 5: Force a Fresh DORA Handshake on the Client
- Once you have identified and corrected the underlying network or server issue, force the client to clear its APIPA state and retry:
- Open the client device's command prompt.
- Run the release command to drop the local APIPA assignment:
- DOS
- ipconfig /release
- Run the renew command to initiate a brand-new DORA handshake:
- DOS
- ipconfig /renew
- Run ipconfig /all to confirm that the client has successfully acquired a valid IP address, subnet mask, and default gateway from your DHCP pool.

Task – 5 Difference between DNS and DHCP

- 1. Name Resolution Failure (Domain Not Found)
- Symptom: nslookup or web browsers return a server failure or state that the domain cannot be found, even though the network link to the DNS server is active.
- Possible Causes: Syntax errors in the BIND zone file, missing trailing dots on Fully Qualified Domain Names (FQDNs), or an un-incremented serial number that prevents resolvers from recognizing zone updates.
- Recommended Solution: Run diagnostic checks like named-checkconf and named-checkzone to pinpoint syntax errors. Correct the records, increment the serial number, and reload the service using rndc reload.
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- 2. Stale Record Resolution (Old IP Still Resolving)
- Symptom: Clients continue to resolve an old IP address long after an A record has been updated on the authoritative server.
- Possible Causes: The record's Time-To-Live (TTL) value was set too high before the change, or intermediate resolvers and local operating systems are caching the old DNS response.
- Recommended Solution: Lower the TTL prior to making planned record changes, or manually flush the client's local DNS cache (e.g., using ipconfig /flushdns on Windows).
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- 3. DNS Query Timeouts / Unresponsive Server
- Symptom: Queries sent to the local DNS server result in DNS request timed out errors.
- Possible Causes: The DNS service daemon (such as BIND9) has crashed or stopped running, or a firewall is actively blocking UDP and TCP port 53 traffic.
- Recommended Solution: Check the service health using `systemctl status bind9`, restart the service, and verify that firewall rules permit inbound traffic on port 53.
- Common DHCP Issues
- 1. APIPA Fallback (169.254.x.x Address)
- Symptom: A client device fails to connect to the network and is assigned an automatic private IP address in the 169.254.x.x range.
- Possible Causes: The DORA handshake failed due to a physical link disconnection, a stopped DHCP service on the server, or complete DHCP address pool exhaustion.
- Recommended Solution: Verify physical and link-layer connectivity, ensure the DHCP server service is running, expand the scope range if exhausted, and force a fresh lease handshake using `ipconfig /renew`.
- 2. Incorrect Default Gateway Assigned
- Symptom: A client successfully receives an IP address and can communicate with other devices on the local subnet, but all external internet and multi-subnet traffic fails.
- Possible Causes: The default gateway field in the DHCP server's active scope configuration was mistyped or misconfigured.
- Recommended Solution: Correct the default gateway IP address within the DHCP server settings, then run `ipconfig /release` followed by `ipconfig /renew` on the client to pull the updated configuration.
- 3. Duplicate IP Address Conflicts
- Symptom: Intermittent network drops occur, and system event logs display duplicate IP address warnings.
- Possible Causes: A static IP address was manually assigned to a server, printer, or network device that falls directly inside the active DHCP dynamic pool range without being excluded.
- Recommended Solution: Configure explicit exclusions in your DHCP server settings for all statically assigned devices, or use DHCP reservations to manage fixed IPs dynamically.
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